Lecture 18: Cubic Spline

Math 6610

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Fall 2026

Today's Goals

- Interpolation error: Interpolation Error I and II, Bounding Lemma, with proofs
- Chebychev nodes, or piecewise polynomials

natural cubic spline

In each interval $[t_i, t_{i+1}]$, $S(x)$ looks like

$$S_i(x) = a_{0,i} + a_{1,i}x + a_{2,i}x^2 + a_{3,i}x^3$$

How do we find $a_{0,i}, a_{1,i}, a_{2,i}, a_{3,i}$ for each i?

Consider knots $t_0, \ldots, t_n$. Follow our example with the following steps:

1. Assume we knew $S''(t_i)$ for each i
2. $S_i''(x)$ is linear, so construct it
3. Get $S_i(x)$ by integrating $S_i''(x)$ twice
4. Impose continuity
5. Differentiate $S_i(x)$ to impose continuity on $S'(x)$

natural cubic spline: Step 1

Assume we knew $S''(t_i)$ for each i

We know $S''(x)$ is continuous. So assume

$$z_i = S''(t_i)$$

(we don't actually know z_i, not yet at least)

natural cubic spline: Step 2

$S_i''(x)$ is linear, so construct it

Since $S_i''(x)$ is linear, and

$$S_i''(t_i) = z_i$$
$$S_i''(t_{i+1}) = z_{i+1}$$

we can write $S_i''(x)$ as

$$\begin{aligned} S_i''(x) &= z_i \frac{t_{i+1} - x}{t_{i+1} - t_i} + z_{i+1} \frac{x - t_i}{t_{i+1} - t_i} \\ &= \frac{z_i}{h_i}(t_{i+1} - x) + \frac{z_{i+1}}{h_i}(x - t_i) \end{aligned}$$

where $h_i = t_{i+1} - t_i$.

natural cubic spline: Step 3

Get $S_i(x)$ by integrating $S_i''(x)$ twice

Take

$$S_i''(x) = \frac{z_i}{h_i}(t_{i+1} - x) + \frac{z_{i+1}}{h_i}(x - t_i)$$

and integrate once:

$$S_i'(x) = -\frac{z_i}{2h_i}(t_{i+1} - x)^2 + \frac{z_{i+1}}{2h_i}(x - t_i)^2 + \hat{C}_i$$

twice:

$$S_i(x) = \frac{z_i}{6h_i}(t_{i+1} - x)^3 + \frac{z_{i+1}}{6h_i}(x - t_i)^3 + \hat{C}_i x + \hat{D}_i$$

adjust:

$$S_i(x) = \frac{z_i}{6h_i}(t_{i+1} - x)^3 + \frac{z_{i+1}}{6h_i}(x - t_i)^3 + C_i(x - t_i) + D_i(t_{i+1} - x)$$

natural cubic spline: Step 4

Impose continuity

For each interval $[t_i, t_{i+1}]$, we require $S_i(t_i) = y_i$ and $S_i(t_{i+1}) = y_{i+1}$:

$$\begin{aligned} y_i = S_i(t_i) &= \frac{z_i}{6h_i}(t_{i+1} - t_i)^3 + \frac{z_{i+1}}{6h_i}(t_i - t_i)^3 + C_i(t_i - t_i) + D_i(t_{i+1} - t_i) \\ &= \frac{z_i}{6}h_i^2 + D_i h_i \\ D_i &= \frac{y_i}{h_i} - \frac{h_i}{6}z_i \end{aligned}$$

and

$$\begin{aligned} y_{i+1} = S_i(t_{i+1}) &= \frac{z_i}{6h_i}(t_{i+1} - t_{i+1})^3 + \frac{z_{i+1}}{6h_i}(t_{i+1} - t_i)^3 + C_i(t_{i+1} - t_i) + D_i(t_{i+1} - t_{i+1}) \\ &= \frac{z_{i+1}}{6}(h_i)^2 + C_i h_i \\ C_i &= \frac{y_{i+1}}{h_i} - \frac{h_i}{6}z_{i+1} \end{aligned}$$

natural cubic spline: Step 4

Impose continuity

So far we have

$$S_i(x) = \frac{z_i}{6h_i}(t_{i+1}-x)^3+\frac{z_{i+1}}{6h_i}(x-t_i)^3+\left(\frac{y_{i+1}}{h_i}-\frac{h_i}{6}z_{i+1}\right)(x-t_i)+\left(\frac{y_i}{h_i}-\frac{h_i}{6}z_i\right)(t_{i+1}-x)$$

natural cubic spline: Step 5

Differentiate $S_i(x)$ to impose continuity on $S'(x)$

$$S_i'(x) = -\frac{z_i}{2h_i}(t_{i+1} - x)^2 + \frac{z_{i+1}}{2h_i}(x - t_i)^2 + \frac{y_{i+1}}{h_i} - \frac{h_i}{6}z_{i+1} - \frac{y_i}{h_i} + \frac{h_i}{6}z_i$$

We need $S_i'(t_i) = S_{i-1}'(t_i)$:

$$S_i'(t_i) = -\frac{h_i}{6}z_{i+1} - \frac{h_i}{3}z_i + \underbrace{\frac{1}{h_i}(y_{i+1} - y_i)}_{b_i}$$

$$S_{i-1}'(t_i) = \frac{h_{i-1}}{6}z_{i-1} + \frac{h_{i-1}}{3}z_i + \underbrace{\frac{1}{h_{i-1}}(y_i - y_{i-1})}_{b_{i-1}}$$

Thus z_i is defined by

$$h_{i-1}z_{i-1} + 2(h_i + h_{i-1})z_i + h_i z_{i+1} = 6(b_i - b_{i-1})$$

natural cubic spline: Step 6

solve

z_i is defined by

$$h_{i-1}z_{i-1} + 2(h_i + h_{i-1})z_i + h_i z_{i+1} = 6(b_i - b_{i-1})$$

- This is $n-1$ equations, $n-1$ unknowns ($z_0 = z_n = 0$ already)
- an $(n-1) \times (n-1)$ tridiagonal system (add 2 for z_0 and z_n)

$$\begin{bmatrix} 1 & & & & & & & \\ h_0 & u_1 & h_1 & & & & & \\ & h_1 & u_2 & h_2 & & & & \\ & & h_2 & u_3 & h_3 & & & \\ & & & \ddots & \ddots & \ddots & & \\ & & & & h_{n-3} & u_{n-2} & h_{n-2} & \\ & & & & & h_{n-2} & u_{n-1} & h_{n-1} \\ & & & & & & & 1 \end{bmatrix} \begin{bmatrix} z_0 \\ z_1 \\ z_2 \\ z_3 \\ \vdots \\ z_{n-2} \\ z_{n-1} \\ z_n \end{bmatrix} = \begin{bmatrix} 0 \\ v_1 \\ v_2 \\ v_3 \\ \vdots \\ v_{n-2} \\ v_{n-1} \\ 0 \end{bmatrix}$$

$$\begin{aligned} u_i &= 2(h_i + h_{i-1}) \\ v_i &= 6(b_i - b_{i-1}) \end{aligned}$$

example

Find the natural cubic spline for

x	-1	0	1
y	1	2	-1

1. Determine h_i, b_i, u_i, v_i

$$h = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad b = \begin{bmatrix} 1 \\ -3 \end{bmatrix} \quad u = \begin{bmatrix} 4 \end{bmatrix} \quad v = \begin{bmatrix} -24 \end{bmatrix}$$

2. Solve

$$\begin{bmatrix} 1 & & \\ 1 & 4 & 1 \\ & & 1 \end{bmatrix} \begin{bmatrix} z_0 \\ z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} 0 \\ -24 \\ 0 \end{bmatrix}$$

3. Result:

$$\begin{bmatrix} z_0 \\ z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} 0 \\ -6 \\ 0 \end{bmatrix}$$

example

Find the natural cubic spline for

x	-1	0	1
y	1	2	-1

1. Plug z_i into

$$\begin{aligned} S_i(x) &= \frac{z_i}{6h_i}(t_{i+1}-x)^3 + \frac{z_{i+1}}{6h_i}(x-t_i)^3 + \left(\frac{y_{i+1}}{h_i} - \frac{h_i}{6}z_{i+1}\right)(x-t_i) \\ &\quad + \left(\frac{y_i}{h_i} - \frac{h_i}{6}z_i\right)(t_{i+1}-x) \end{aligned}$$

$$S(x) = \begin{cases} -(x+1)^3 + 3(x+1) - x & -1 \leqslant x < 0 \\ -(1-x)^3 - x + 3(1-x) & 0 \leqslant x < 1 \end{cases}$$

natural cubic spline: the reduced system

The equation of Step 5 holds at the interior knots, $i = 1, \ldots, n-1$. With $z_0 = z_n = 0$ the terms $h_0 z_0$ and $h_{n-1} z_n$ drop out. For $n \geqslant 2$, the middle $n-1$ rows of Step 6 are

$$\underbrace{\begin{bmatrix} u_1 & h_1 & & \\ h_1 & u_2 & \ddots & \\ & \ddots & \ddots & h_{n-2} \\ & & h_{n-2} & u_{n-1} \end{bmatrix}}_{=:A} \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_{n-1} \end{bmatrix} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_{n-1} \end{bmatrix}.$$

- A is $(n-1) \times (n-1)$, symmetric, tridiagonal, positive entries.
- For $n = 1$ there is no equation: $z_0 = z_1 = 0$, and S is the line through the two points.

strictly diagonally dominant

definition

A matrix $A \in \mathbb{R}^{m \times m}$ is *strictly diagonally dominant* (by rows) if $|a_{ii}| > \sum_{k \neq i} |a_{ik}|$ for every i.

Our A: in row i the off-diagonal sum is at most

$$h_{i-1} + h_i < 2(h_{i-1} + h_i) = u_i,$$

since all $h_i > 0$. So A is strictly diagonally dominant.

The Natural Cubic Spline Is Unique

Proposition

A strictly diagonally dominant matrix is nonsingular. Hence the natural cubic spline exists and is unique (for every $n \geqslant 1$ and all data).

Proof. Idea: look at the largest entry of a null vector.
1. Row i. Let $Aw = 0$ and $|w_i| = \max_k |w_k|$. Row i gives $a_{ii}w_i = -\sum_{k\neq i} a_{ik}w_k$, so

$$|a_{ii}|\,|w_i| \leqslant \sum_{k\neq i} |a_{ik}|\,|w_k| \leqslant \left(\sum_{k\neq i} |a_{ik}|\right)|w_i|\,.$$

2. Strict dominance then forces $|w_i| = 0$, so $w = 0$: A is nonsingular.
3. The spline. By Steps 1–5, S is a natural cubic spline through the data exactly when S_i is the formula of Step 4 with $z_0 = z_n = 0$ and $Az = v$. This has exactly one solution z, so there is exactly one S. ■

No Pivoting Needed

Proposition

Gaussian elimination without pivoting on A never meets a zero pivot, all multipliers satisfy $0 < l_i < 1$, and there is no growth: $\rho \leqslant 1$.

Proof. Idea: each pivot stays larger than the entry next to it.
Write d_i diagonal, $e_i > 0$ off-diagonal ($d_i = u_i$, $e_i = h_i$), pivots $q_1 = d_1$, $q_i = d_i - l_i e_{i-1}$, multipliers $l_i = e_{i-1}/q_{i-1}$; $m = n - 1$ rows.
1. Claim $q_i > e_i$ ($e_m := 0$), by induction on i. Base: $q_1 = d_1 > e_1$. Step: $q_{i-1} > e_{i-1}$ gives $0 < l_i < 1$, so by dominance of row i

$$q_i > d_i - e_{i-1} > e_i.$$

2. Growth. U has entries q_i and e_i, and $0 < q_i \leqslant d_i$, so $\rho \leqslant 1$. ■

At step i the only entry below the pivot is $e_i < q_i$: partial pivoting would make no swaps here.

example 2

Find the natural cubic spline for

x	0	1	2
y	0	1	0

1. $h = (1, 1),\ b = (1, -1),\ u_1 = 4,\ v_1 = 6(-1 - 1) = -12$
2. Solve $4z_1 = -12$: $z = (0, -3, 0)$
3. Plug z_i into $S_i(x)$ of Step 4 ($C_0 = \frac{3}{2}$, $D_0 = 0$, $C_1 = 0$, $D_1 = \frac{3}{2}$) and expand:

$$S(x) = \begin{cases} \frac{3}{2}x - \frac{1}{2}x^3, & 0 \leqslant x \leqslant 1, \\ 1 - \frac{3}{2}(x-1)^2 + \frac{1}{2}(x-1)^3, & 1 \leqslant x \leqslant 2. \end{cases}$$

Check: $S_0(1) = 1 = S_1(1)$; $S_0'(1) = \frac{3}{2} - \frac{3}{2} = 0 = S_1'(1)$; $S_0''(1) = -3 = S_1''(1)$; $S_0''(0) = 0$, $S_1''(2) = -3 + 3 = 0$; $S_1(2) = 1 - \frac{3}{2} + \frac{1}{2} = 0$.

natural cubic spline:

Second route: in each interval write S_i in powers of $s = x - t_i$,

$$S_i(x) = a_{0,i} + a_{1,i}s + a_{2,i}s^2 + a_{3,i}s^3 = a_{0,i} + s\big(a_{1,i} + s(a_{2,i} + s\,a_{3,i})\big).$$

$S_i''(t_i) = 2a_{2,i}$, so $a_{2,i} = z_i/2$; also set $a_{2,n} := z_n/2$. Constraints:

1. $S_i(t_i) = y_i$, $S_i(t_{i+1}) = y_{i+1}$, $\ i = 0, \ldots, n-1$
2. $S_{i-1}'(t_i) = S_i'(t_i)$, $\ i = 1, \ldots, n-1$
3. $S_{i-1}''(t_i) = S_i''(t_i)$, $\ i = 1, \ldots, n-1$
4. natural: $a_{2,0} = a_{2,n} = 0$

From 3: $2a_{2,i} + 6a_{3,i}h_i = 2a_{2,i+1}$. From 1: $a_{0,i} = y_i$ and $y_{i+1} - y_i = a_{1,i}h_i + a_{2,i}h_i^2 + a_{3,i}h_i^3$. So

$$a_{3,i} = \frac{a_{2,i+1} - a_{2,i}}{3h_i}, \qquad a_{1,i} = b_i - \frac{h_i}{3}(2a_{2,i} + a_{2,i+1}).$$

natural cubic spline

From 2: insert $a_{3,i-1}$, then $a_{1,i-1}$ and $a_{1,i}$:

$$\begin{aligned} S'_{i-1}(t_i) &= a_{1,i-1} + 2a_{2,i-1}h_{i-1} + 3a_{3,i-1}h_{i-1}^2 = a_{1,i-1} + h_{i-1}(a_{2,i-1} + a_{2,i}) \\ &= b_{i-1} + \frac{h_{i-1}}{3}(a_{2,i-1} + 2a_{2,i}), \\ S'_i(t_i) &= a_{1,i} = b_i - \frac{h_i}{3}(2a_{2,i} + a_{2,i+1}). \end{aligned}$$

Setting them equal and multiplying by 3: for $i = 1, \ldots, n-1$,

$$h_{i-1}a_{2,i-1} + 2(h_{i-1} + h_i)a_{2,i} + h_i a_{2,i+1} = 3(b_i - b_{i-1}).$$

- With $a_{2,i} = z_i/2$ this is the equation of Step 5 divided by 2: the same matrix A.
- Nested-form coefficients, used by the Matlab code below: $a_{0,i} = y_i$, $a_{1,i} = b_i - \frac{h_i}{6}(2z_i + z_{i+1})$, $a_{2,i} = \frac{z_i}{2}$, $a_{3,i} = \frac{z_{i+1} - z_i}{6h_i}$.

Algorithm

1. Compute for $i = 0, \ldots, n-1$

$$h_i = t_{i+1} - t_i \qquad b_i = \frac{1}{h_i}(y_{i+1} - y_i)$$

2. Set u, v:
3. tridiagonal solve to get z
4. substitute into the nested form for $S(x)$

Algorithm: Matlab

Solve $Az = v$ by elimination without pivoting ("Thomas algorithm"):

```
function pp = natspline(t, y)
% natural cubic spline; evaluate with ppval(pp, x)
t = t(:); y = y(:); n = length(t) - 1;
h = diff(t);  b = diff(y)./h;       % h_i, b_i, i = 0..n-1
z = zeros(n+1, 1);                  % z_0..z_n, z_0 = z_n = 0
if n >= 2
  u = 2*(h(1:n-1) + h(2:n));        % u_1..u_{n-1} -> pivots
  v = 6*(b(2:n) - b(1:n-1));        % v_1..v_{n-1}
  for i = 2:n-1                     % forward sweep
    l = h(i)/u(i-1);
    u(i) = u(i) - l*h(i);
    v(i) = v(i) - l*v(i-1);
  end
  z(n) = v(n-1)/u(n-1);             % back substitution
  for i = n-2:-1:1
    z(i+1) = (v(i) - h(i+1)*z(i+2))/u(i);
  end
end
```

Algorithm: Matlab

Nested-form coefficients $a_{3,i}, a_{2,i}, a_{1,i}, a_{0,i}$, stored as a Matlab piecewise polynomial:

```
19 a3 = (z(2:n+1) - z(1:n))./(6*h);          % s^3
20 a2 = z(1:n)/2;                            % s^2
21 a1 = b - h.*(2*z(1:n) + z(2:n+1))/6;      % s^1
22 pp = mkpp(t', [a3 a2 a1 y(1:n)]);         % highest power first
```

- Cost: forward sweep 5 flops per row (one division, two multiply-subtracts), back substitution 3 per row: $\approx 8n$ flops, $O(n)$, versus $\approx \frac{2}{3}n^3$ for dense LU.
- Test: `natspline([-1 0 1], [1 2 -1])` gives $z = (0, -6, 0)$ (example); `natspline([0 1 2], [0 1 0])` gives $z = (0, -3, 0)$ (example 2).

Why the System Is So Nice

- **unique spline:** strictly diagonally dominant $\Rightarrow A$ nonsingular
- **no pivoting needed:** pivots stay positive, multipliers < 1, $\rho \leqslant 1$; by Lecture 16, elimination is backward stable here
- $O(n)$ **solve:** tridiagonal LU, $\approx 8n$ flops. A spline through 10^6 points costs about 10^7 flops: a fraction of a second.

Other end conditions:

- fixed-slope: z_0, z_n become unknowns, with two more equations from $S'(t_0) = a$, $S'(t_n) = b$; again tridiagonal and strictly diagonally dominant
- periodic: a "cyclic" tridiagonal matrix; not-a-knot: nearly tridiagonal

Symmetric, tridiagonal, diagonally dominant matrices come back in finite differences for ODE boundary value problems.

cubic spline: which end condition?

end condition	needs	Matlab
not-a-knot	data only	`spline(t,y)` (default)
fixed-slope	end slopes a, b	`spline(t,[a y b])`
natural	data only	`csape(t,y,'variational')`
periodic	$y_0 = y_n$	`csape(t,y,'periodic')`

(`csape`: Curve Fitting Toolbox. Evaluate with `ppval(pp,x)`.)

- **not-a-knot:** uses no extra information; good default
- **fixed-slope:** use when the true end slopes are known (e.g. a velocity at the endpoints); then it is the most accurate (later today)
- **natural:** simplest to analyze, but it forces $S'' = 0$ at the ends; if $f''(t_0) \neq 0$ or $f''(t_n) \neq 0$, accuracy near the ends drops (later today)
- **periodic:** closed curves, periodic signals (angles, daily cycles)

The Smoothest Interpolant

definition

For g continuous on $[a,b]$, the L^2 norm is

$$\|g\|_2 = \left(\int_a^b |g(x)|^2\,dx\right)^{1/2} \qquad \text{(the continuous analogue of the vector 2-norm)}.$$

Theorem 17.7: Minimum Curvature

Let $S(x)$ be a cubic spline with $S(t_i) = y_i$, $i = 0,\ldots,n$. Let f have f'' continuous on $[a,b]$ and $f(t_i) = y_i$, $i = 0,\ldots,n$. If

$$S''(x)\big[f'(x) - S'(x)\big]\Big|_{x=a}^{x=b} = 0 \qquad \text{(boundary identity)},$$

then $\|S''\|_2 \leqslant \|f''\|_2$, with equality only if $f = S$.

Among all such curves through the data, the cubic spline has the least total squared curvature $\int_a^b (f'')^2\,dx$: it is the "flattest". Proof: next slides.

When Does the Boundary Identity Hold?

Proposition

$S''(f' - S')\big|_a^b = 0$ holds in these cases:

- S natural, any f: $S''(a) = S''(b) = 0$.
- S fixed-slope, f with the same end slopes $f'(a) = S'(a), f'(b) = S'(b)$: $f' - S' = 0$ at both ends.
- S periodic, f with $f'(a) = f'(b)$: using $S''(a) = S''(b)$ and $S'(a) = S'(b)$,

$$S''(b)\big[f'(b) - S'(b)\big] - S''(a)\big[f'(a) - S'(a)\big] = S''(a)\big[f'(b) - f'(a)\big] = 0.$$

- not-a-knot: neither S'' nor $f' - S'$ need vanish at the ends; Theorem 17.7 is not guaranteed.
- natural cubic spline: flattest among *all* interpolants. Fixed-slope: flattest among interpolants with the same end slopes.

Why “Spline”?

A draftsman’s spline is a thin flexible strip held by weights at the data points. Its bending energy is proportional to

$$\int \kappa_c^2 \, ds, \qquad \kappa_c = \frac{f''}{(1 + (f')^2)^{3/2}}, \quad ds = \sqrt{1 + (f')^2}\, dx$$

(curvature of the graph, integrated over arc length).

- For small slopes, $|f'| \ll 1$, this is about $\int_a^b (f'')^2 \, dx$: the quantity minimized in Theorem 17.7.
- The free ends of the strip carry no bending moment: $S'' = 0$, the natural cubic spline.
- The proof (next slide) is one integration by parts plus Pythagoras: $f'' - S''$ is orthogonal to S''. The same idea is behind least squares and finite element methods.

Proof of Theorem 17.7

Proof. Idea: show $S'' \perp (f'' - S'')$, then use Pythagoras.

1. *One interval* $[t_i, t_{i+1}]$: $S''' = c_i$, $f - S = 0$ at the ends. By parts:

$$\int_{t_i}^{t_{i+1}} S''(f'' - S'')\,dx = \Big[S''(f' - S')\Big]_{t_i}^{t_{i+1}} - c_i \underbrace{\big[f - S\big]_{t_i}^{t_{i+1}}}_{=0}.$$

2. *Sum over* i. The brackets telescope (S'', $f' - S'$ continuous):

$$\int_a^b S''(f'' - S'')\,dx = \Big[S''(f' - S')\Big]_a^b = 0.$$

3. *Pythagoras.* With $g := f'' - S''$, square $f'' = S'' + g$ and integrate:

$$\|f''\|_2^2 = \|S''\|_2^2 + 2\underbrace{\int_a^b S''g\,dx}_{=0} + \|g\|_2^2 \;\geqslant\; \|S''\|_2^2.$$

4. *Equality:* $\|g\|_2 = 0$, g continuous, so $g \equiv 0$: $f - S$ is linear, 0 at a, b; $f = S$. ■

Theorem 17.7: The Geometry

Step 2 says S'' is orthogonal to the error $f'' - S''$ in the inner product $\langle u, v \rangle = \int_a^b uv\,dx$:

$$\underbrace{\|f''\|_2^2}_{\text{hypotenuse}^2} = \underbrace{\|S''\|_2^2}_{\text{leg}^2} + \underbrace{\|f'' - S''\|_2^2}_{\text{leg}^2}.$$

Where each assumption is used:

- S cubic on each interval: S''' constant there (Step 1)
- f, S interpolate the same data: $[f - S] = 0$ (Step 1)
- S'' and f'' continuous: telescoping (Step 2)
- boundary identity: the last bracket is 0 (Step 2)

Some analysis...

what can we say about

$$e(t) = f(t) - p_n(t)$$

at some point x? Consider $p = 1$: linear interpolation of a function at $x = x_0, x_1$

- want: error at x, $e(x)$
- look at

$$g(t) = e(t) - \frac{(t - x_0)(t - x_1)}{(x - x_0)(x - x_1)} e(x)$$

- $g(t)$ is 0 at $t = x_0, x_1, x$
- so $g'(t)$ is zero at two points
- so $g''(t)$ is zero at one point, call it c
- then

$$\begin{aligned} 0 = g''(c) &= e''(t) - 2\frac{e(x)}{(x - x_0)(x - x_1)} \\ &= f''(t) - 2\frac{e(x)}{(x - x_0)(x - x_1)} \\ e(x) &= \frac{(x - x_0)(x - x_1)}{2} f''(c) \end{aligned}$$

Theorem: Interpolation Error I

If $p_n(x)$ is the (at most) n degree polynomial interpolating $f(x)$ at $n+1$ distinct points and if $f^{(n+1)}$ is continuous, then

$$e(x) = f(x) - p_n(x) = \frac{1}{(n+1)!} f^{(n+1)}(c) \prod_{i=0}^{n} (x - x_i)$$

Theorem: Bounding Lemma

Suppose x_i are equispaced in $[a, b]$ for $i = 0, \ldots, n$. Then

$$\prod_{i=0}^{n} |x - x_i| \leqslant \frac{h^{n+1}}{4} n!$$

Theorem: Interpolation Error II

Let $|f^{(n+1)}(x)| \leqslant M$, then with the above,

$$|f(x) - p_n(x)| \leqslant \frac{M h^{n+1}}{4(n+1)}$$

Interpolation Error I: Proof

Precisely: $f^{(n+1)}$ continuous on $[a,b]$, nodes $x_0 < \cdots < x_n$ in $[a,b]$, $x \in [a,b]$; then $c = c(x) \in (a,b)$.

Proof. Idea: as for $n = 1$: a function with $n+2$ zeros, then Rolle $n+1$ times. Write $\omega(t) := \prod_{i=0}^{n}(t - x_i)$.

1. Setup. If x is a node, both sides are 0. Otherwise $\omega(x) \neq 0$; set

$$g(t) := e(t) - \frac{\omega(t)}{\omega(x)}\, e(x), \qquad t \in [a,b].$$

2. Zeros. $g = 0$ at $x_0, \ldots, x_n$ and at x: $n+2$ distinct zeros in $[a,b]$.

3. Rolle. Between two zeros of a function lies a zero of its derivative. So g' has $n+1$ distinct zeros, $\ldots$, $g^{(n+1)}$ has one, $c \in (a,b)$.

4. Differentiate $n+1$ times. $p_n^{(n+1)} = 0$ (degree $\leqslant n$) and $\omega^{(n+1)} = (n+1)!$ ($\omega = t^{n+1} + \ldots$):

$$0 = g^{(n+1)}(c) = f^{(n+1)}(c) - (n+1)!\,\frac{e(x)}{\omega(x)} \implies e(x) = \frac{f^{(n+1)}(c)}{(n+1)!}\,\omega(x).$$

■

Bounding Lemma: Proof

Precisely: $h = (b-a)/n$, $x_i = a + ih$, $i = 0, \ldots, n$; any $x \in [a, b]$.

Proof. 1. Locate x. Choose $j \in \{0, \ldots, n-1\}$ with $x \in [x_j, x_{j+1}]$.
2. Nearest nodes. Two distances with sum h: $\left|(x - x_j)(x - x_{j+1})\right| \leqslant (h/2)^2$.
3. Other nodes. $|x - x_i| \leqslant (j + 1 - i)h$ for $i < j$, and $|x - x_i| \leqslant (i - j)h$ for $i > j + 1$. Multiply all $n + 1$ factors:

$$\prod_{i=0}^{n} |x - x_i| \leqslant \frac{h^2}{4} \cdot (j+1)!\, h^j \cdot (n-j)!\, h^{n-j-1} = \frac{h^{n+1}}{4}\, (j+1)!\, (n-j)!.$$

4. Factorials. $1 \leqslant j + 1 \leqslant n$, so $\binom{n+1}{j+1} \geqslant n + 1$ and $(j+1)!\,(n-j)! = (n+1)!/\binom{n+1}{j+1} \leqslant n!$. ■

Interpolation Error II follows: with $\left|f^{(n+1)}\right| \leqslant M$ on $[a, b]$, $|e(x)| \leqslant \frac{M}{(n+1)!} \cdot \frac{h^{n+1}}{4} n! = \frac{Mh^{n+1}}{4(n+1)}$.

Reading the Error Formula

$$e(x) = \underbrace{\frac{f^{(n+1)}(c)}{(n+1)!}}_{\text{how smooth } f \text{ is}} \underbrace{\prod_{i=0}^{n}(x - x_i)}_{\text{where the nodes are}}$$

- Bound: with $\left|f^{(n+1)}\right| \leqslant M$ on $[a, b]$,

$$\max_{[a,b]} |e(x)| \leqslant \frac{M}{(n+1)!} \max_{x \in [a,b]} \prod_{i=0}^{n} |x - x_i| .$$

  (M exists: $f^{(n+1)}$ is continuous on the closed interval.)
- Only the second factor is ours to choose: the nodes.
- Equally spaced: the product is largest in the first and last intervals (plot on the Chebychev slide below).

Fixes

We have two options:

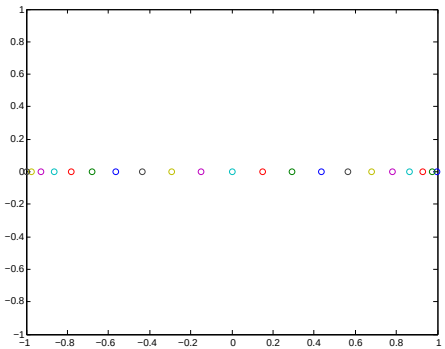
1. move the nodes: Chebychev nodes
2. piecewise polynomials (splines)

Option #1: Chebychev nodes in $[-1, 1]$

$$x_i = cos(\pi \frac{2i+1}{2n+2}), \quad i = 0, \ldots, n$$

Option #2: piecewise polynomials...

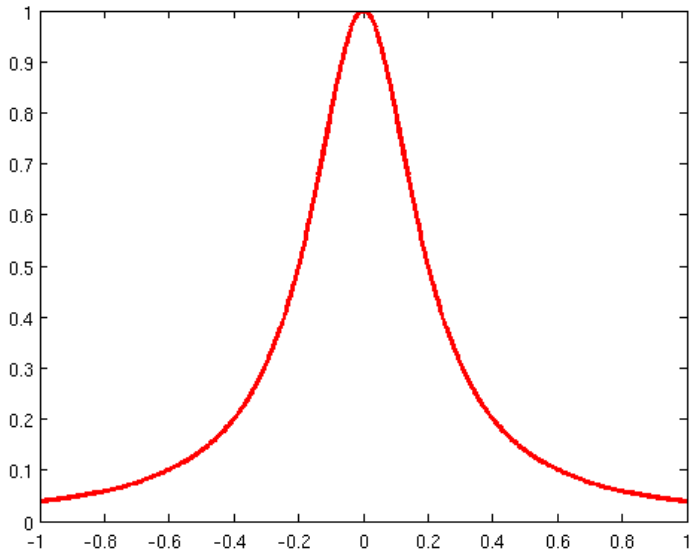
Chebychev Nodes



- Can obtain nodes from equidistant points on a circle projected down
- Nodes are non uniform and non nested

Chebychev Nodes

High degree polynomials using equispaced points suffer from many oscillations



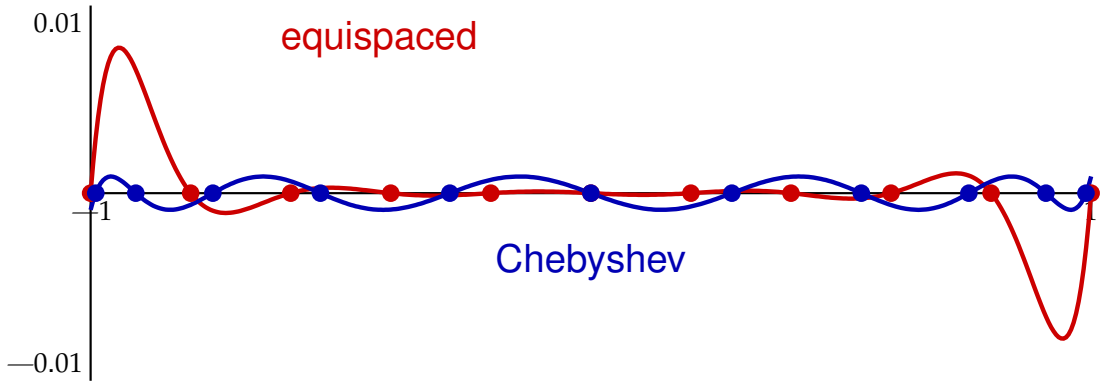
- Points are bunched at the ends of the interval
- Error is distributed more evenly

Chebychev Nodes: The Product

Proposition (proof omitted)

For the Chebychev nodes $x_i = \cos(\pi \frac{2i+1}{2n+2})$, $i = 0, \ldots, n$, in $[-1, 1]$: $\max_{[-1,1]} \prod_i |x - x_i| = 2^{-n}$, and every other choice of $n+1$ nodes in $[-1, 1]$ gives at least 2^{-n}. (On $[a, b]$: map $x \mapsto \frac{a+b}{2} + \frac{b-a}{2} x$.)

Example. $\prod_i (x - x_i)$ for $n = 10$:



Equally spaced: max $\approx 8.5 \cdot 10^{-3}$, large near the ends. Chebychev: $2^{-10} \approx 9.8 \cdot 10^{-4}$, even ripples.

Runge's Example Explained

Example. $f(x) = 1/(1+25x^2)$ on $[-1,1]$; max error (computed):

n	5	10	20
equally spaced	0.43	1.92	59.8
Chebychev	0.56	0.11	0.015

- The smoothness factor is large: $f = \sum_{k\geqslant 0}(-25x^2)^k$ for $|x| < 1/5$, so $\left|f^{(m)}(0)\right|/m! = 5^m$ for even m; so $M/(n+1)! \geqslant 5^{n+1}$ for odd n.
- Equally spaced: the product is too large near the ends; the error there grows with n.
- Chebychev: the bound is $\geqslant 5^{n+1}2^{-n}$ and grows, yet the observed error decreases. The bound is only an upper bound. (Fact, proof omitted: for every Lipschitz continuous f, Chebychev interpolants converge uniformly.)
- Rule: high degree $\Rightarrow$ Chebychev nodes; fixed grid $\Rightarrow$ splines.

Error of a degree 1 spline

Corollary

Let f'' be continuous on $[a,b]$, $S(x)$ the degree 1 spline with $S(t_i) = f(t_i)$, $i = 0, \ldots, n$, and $h := \max_i h_i$. Then

$$\max_{x\in[a,b]} |f(x) - S(x)| \leqslant \frac{h^2}{8} \max_{[a,b]} |f''| \qquad (O(h^2)\text{: halving } h \text{ divides the bound by } 4).$$

Proof. 1. One interval. For $x \in [t_i, t_{i+1}]$, S is the line through f at t_i, t_{i+1}. Interpolation Error I with $n = 1$: for some $c \in (t_i, t_{i+1})$,

$$f(x) - S(x) = \frac{f''(c)}{2}(x - t_i)(x - t_{i+1}).$$

2. The product is largest at the midpoint: $|(x - t_i)(x - t_{i+1})| \leqslant (h_i/2)^2 \leqslant h^2/4$.

3. Combine: $|f(x) - S(x)| \leqslant \frac{1}{2} \cdot \frac{h^2}{4} \max |f''|$. ■

Error of a cubic spline

Theorem (Hall–Meyer 1976; proof omitted)

Let $f^{(4)}$ be continuous on $[a,b]$, and $S(x)$ the fixed-slope cubic spline with $S(t_i) = f(t_i)$ and end slopes $f'(t_0), f'(t_n)$. With $h := \max_i h_i$,

$$\max_{[a,b]} |f - S| \leqslant \frac{5}{384} h^4 \max_{[a,b]} \left| f^{(4)} \right| .$$

Example. $f = \sin$ on $[0, \pi/2]$, n equal intervals; max error (computed):

n	4	8	16	32	ratio
degree 1	$1.9 \cdot 10^{-2}$	$4.8 \cdot 10^{-3}$	$1.2 \cdot 10^{-3}$	$3.0 \cdot 10^{-4}$	4
natural	$7.7 \cdot 10^{-3}$	$1.9 \cdot 10^{-3}$	$4.7 \cdot 10^{-4}$	$1.2 \cdot 10^{-4}$	4
fixed-slope	$6.3 \cdot 10^{-5}$	$3.9 \cdot 10^{-6}$	$2.4 \cdot 10^{-7}$	$1.5 \cdot 10^{-8}$	16

(Check, $n = 4$: $h^2/8 = 0.019$ and $\frac{5}{384} h^4 = 3.1 \cdot 10^{-4}$.)

Natural: only $O(h^2)$ here: it forces $S''(\pi/2) = 0$, but $\sin''(\pi/2) = -1$.

Next time...

- B-splines